Homework 3

Foundations of Quantum Optics

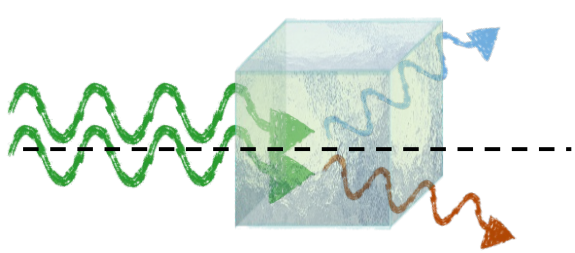
OPTI 544, Spring 2026

1. **Four-wave mixing:** Consider a third-order nonlinear medium with induced nonlinear polarization as follows:

$$\vec{P}_i^{NL} = \epsilon_0 \chi^{(3)}_{ijkl} \vec{E}_j \vec{E}_k \vec{E}_l, \tag{1}$$

where $i = x, y, z$ is the direction for the induced polarization, and $j, k, l = x, y, z$ are the directions corresponding to the polarization of the electric field. For simplicity we will first consider that only χ_{zzzz} is non-zero, and all the other χ_{ijkl} vanish.

(a) Consider a green laser incident on such a medium. The nonlinearity of the medium can lead to a process where two green photons can combine to generate a pair of entangled red and blue photons as follows:



Simplify the following nonlinear Hamiltonian to obtain the energy and momentum conservation conditions for the above process:

$$\hat{H}^{(3)}_{NL} = -\int_V dV \epsilon_0 \chi^{(3)}_{zzzz} \vec{E}_z \vec{E}_z \hat{\vec{E}}_z \hat{\vec{E}}_z. \tag{2}$$

The electric field corresponding to the green laser is $\vec{E}_z = \vec{z}\left[\mathcal{E}_G e^{-i(\omega_G t - kx)} + \mathcal{E}_G^* e^{i(\omega_G t - kx)}\right]$, the red and blue electric fields correspond to the 'signal' and 'idler' modes generated as a result of the process.

(b) What is the output state for the red and blue modes, considering that the initial state of the field in all modes other than the pump is in vacuum and the nonlinearity is weak?

(c) Considering a nonlinear medium with nonzero $\chi^{(3)}_{zzzz}$ and $\chi^{(3)}_{zzyy}$ (all other $\chi^{(3)}_{ijkl} = 0$), using the simplified Hamiltonian from part (a), show that four-wave mixing can generate two polarization entangled photons.

2. **Thermal state**: Consider the thermal density matrix of a single-mode field described by:

$$\rho_{\text{th}} \equiv \sum_n P_n \left|n\right\rangle \left\langle n\right| ,$$

where the probability of having n photons is given by the Boltzmann distribution

$$P_n = \frac{e^{-(n+1/2)\hbar\omega/(k_B T)}}{\sum_{m=0}^{\infty} e^{-(m+1/2)\hbar\omega/(k_B T)}}.$$

(a) What is the expectation value of the number of photons in a thermal state $\langle \hat{n} \rangle \equiv \text{Tr}\left[\hat{n}\rho_{\text{th}}\right]$ as a function of the temperature and frequency of the mode?

Hint: In order to evaluate the sum $f(x) \equiv \sum_{n=0}^{\infty} n e^{-nx}$, use first the fact that $g(x) = \sum_{n=0}^{\infty} e^{-nx} = \frac{1}{1-e^{-x}}$ (geometric series sum); and then note that $f(x) = \frac{dg(x)}{dx}$.

(b) What is the average number of photons around visible wavelengths ~ 500 nm at room temperature ~ 300 K and temperature of the sun ~ 6000 K?